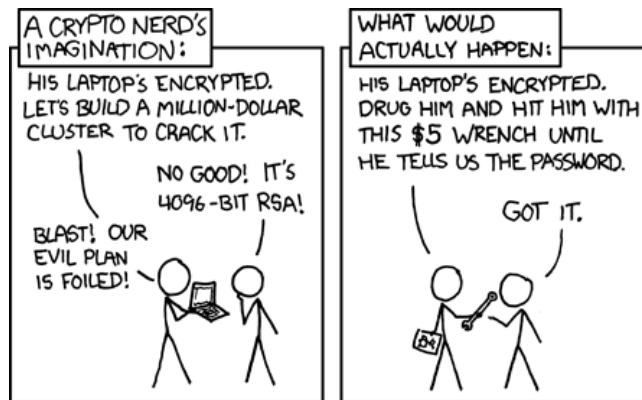


CRYPTANALYSIS: HISTORICAL CIPHERS

Lab 1 in TSIT03 Cryptography,
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– <http://xkcd.com/538/>

1 Introduction

In this laboratory work you will try to decipher three different ciphers: transposition, simple substitution and Vigenère ciphers. You have no knowledge of which method or key is used on your texts, except that you have one of each method. Before the laboratory you should read the instruction and the part of the course literature dealing with elementary ciphers. This will make the work easier. You have two hours scheduled with an assistant. Should you not finish in these two hours you can continue by yourself whenever a computer is available and report to the assistant later. If you have to use more than four hours to decipher your texts you should contact the assistant. Maybe you got a difficult text.

Do *not* hesitate to ask the assistant for help when you are stuck!

2 Goals of the lab

The goal is simple: perform a successful cryptanalysis and find the plaintexts!

3 Preparing for the lab

Before the lab you should have read up on the methods for breaking the historic ciphers as given in the lecture. It is important that you understand how the transposition, substitution, and the Vigenère ciphers work, and that you have a basic idea of their weaknesses.

4 The software Knekt

The software available to you is called KNEKT, and its main functionality is to perform frequency analysis on the given ciphers.

You can access the program at <https://icg.gitlab-pages.liu.se/crypto/knekt/>

5 Performing the laboration

For each ciphertext given (as a number) to you by the assistant, you are supposed to

1. Find out (make an educated guess) which cipher method has been used (Substitution, Transposition or Vigenère).
2. Break the cipher and find the plaintext. Hint: all the texts are in English.

You load your ciphertext (the given number) in the drop-down menu to the left in the web application.

If you are doing the lab on your own, solve your assigned ciphertexts. If you have a lab partner (or several), solve the ciphertexts assigned to the person whose last name comes first in alphabetical order.

Hint: When you are solving a Vigenère cipher you will need the Bigram Distance Counter. This table shows the number of bigrams at each distance. The number of bigrams (count) at each distance is shown below the distance 2, 3, 4, and so on. It's good to know that the bigram distance 2 will be very common because of language, therefore this distance can be ignored.

Important hint: There is a structure to the solution (i.e. the alphabetical permutation) for the substitution ciphers. A key phrase displaces the first few characters of the alphabet, then the rest appear as normal. So for instance, if the key phrase would be KEY, the solution for the cipher would be KEYABCFGHI-JLMNOPQRSTUVWXYZ (so A encodes to K, B encodes to E, C encodes to Y, D encodes to A etc). The key is entered or read under the alphabet in the "Plain alphabet" part of the Knekt substitution cipher tool. Ask if this is unclear.

Good luck, have fun!

6 Handing in your results

Once you've cracked your assigned ciphers, you hand in the results on the coursepage in Lisam, under Submissions → Lab 1 assignment. **If you had a lab partner, add them to the group before handing in.** Remember that you will have to show your results to one of the lab assistants during the lab in addition to handing in your results on Lisam.

Either attach a document (PDF, Word, txt, md) or simply write directly in the text field. Include the following:

1. Which ciphers you were assigned, i.e. XXX.cry
2. For each cipher, specify the type you found it to be: Substitution, transposition or Vigenère
3. For each cipher, specify what the key was: For substitution and Vigenère ciphers, this will be a short key phrase, whilst for transposition ciphers it will be the ordering of the columns